Yilin (Elsa) Wang

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SUMMARY

Applied AI and workflow automation engineer with 2+ years of enterprise SaaS support experience and procurement domain depth. Builds Python systems where LLMs and tool-calling agents handle messy operational workflows under structured outputs, guardrails, evaluation harnesses, and durable human-approval checkpoints (LangGraph). Has worked directly with development, QA, product, and business teams to diagnose issues and turn operational pain points into requirements for tested, reviewable systems.

SKILLS

Applied AI Engineering: deterministic-first parsing, validation rules, structured LLM outputs, LLM-assisted extraction, prompt design, multi-step agents & tool calling, agent orchestration (LangGraph), document extraction (Docling), semantic search / embeddings, human-in-the-loop review, guardrails, audit trails, evaluation harnesses, tracing and observability
Core Technical Skills: Python, FastAPI, Pydantic, Anthropic Claude API, sentence-transformers, SQL, SQLAlchemy, SQLite, pytest, REST APIs, JSON, Git/GitHub, CI/CD (GitHub Actions), Docker, Kubernetes (local), MS SQL Server, Oracle, HTML/CSS/JavaScript/TypeScript fundamentals
Enterprise Systems & Support: Jira, Salesforce, SAP Ariba, Postman, Harness exposure, Linux environments, troubleshooting, root cause analysis, defect reproduction, stakeholder communication

PROJECT

RFQ AI Workflow Automation Project - Human-Supervised, Deterministic-First Procurement AI | Feb 2026 - Present
Public GitHub repo; built independently with synthetic/mock RFQ data — no confidential client, supplier, or pricing data

- Built a modular Python/FastAPI system that parses SAP Ariba-style Excel RFQs into validated JSON, with a provider-neutral Claude analyzer (Pydantic-validated, human_review_required guardrail) that reached 91% field accuracy vs. 46% for the deterministic baseline on an 11-case hand-labeled eval, with cost and latency measured; added local semantic supplier search (100% precision/recall, 9-case eval) and Docling quotation extraction (8/8, 8-case eval).
- Built a multi-step, tool-calling agent (Claude Sonnet) with a hard iteration cap and full tool-call tracing, then rebuilt it in LangGraph with a human-approval pause: drafted supplier emails stop for approve or reject-with-note, state survives restarts via a SQLite checkpointer, and sending stays outside the graph; parity with the original loop verified by 19 mutation-tested tests and a 12-call live comparison.
- Backed the system with 147 pytest tests (proven to fail when code is deliberately broken), GitHub Actions CI, and one trace_id across logs, checkpoints, and reviews; live testing surfaced real bugs (SQLite thread-safety, answer parsing, missing quantity). Containerized with Docker (durable pause verified across a recreated container) and deployed to a local kind Kubernetes cluster with five core endpoints verified.

WORK EXPERIENCE

DECI Limited - Procurement Engineer, Limerick, Ireland | Jun 2025 - Jun 2026

- Owned end-to-end RFQ sourcing workflows for 8 client accounts, processing about 200 RFQs per month through SAP Ariba, Excel, and SharePoint, and structuring quotation evidence and missing-field follow-up to speed downstream QC and management review.
- Identified over 20 new suppliers for niche, hard-to-source industrial items, resolving technical queries across manufacturers, distributors, and Middle East end users on specifications, certificates, pricing, and lead times.
- Mentored 2 interns on the sourcing workflow and translated recurring RFQ pain points into the validation, vendor-matching, auditability, and human-review requirements that seeded the RFQ AI Workflow Automation project.

Kaleris (formerly Navis) — Associate Support Engineer, United States | Feb 2021 - May 2023

- Delivered customer-facing Tier-2 technical support for container terminal operating system (TOS) SaaS applications, managing over 45 customer cases per month with less than 10% escalation rate by troubleshooting application issues, verifying data with SQL, reproducing defects, and documenting root cause analysis in Salesforce CRM.
- Coordinated cross-functional Jira-based escalations across four teams (development, QA, product, and regional support) to isolate defects, clarify customer impact, and drive issues from investigation to resolution while supporting SLA adherence and clear customer communication.
- Built support knowledge base content — detailed case documentation, reproducible issue notes, and solution guidance across Windows and Linux environments — improving repeat-issue handling and operational reliability during regular support and on-call coverage.

EDUCATION

University of Limerick — MSc Software Development: International Systems, **First Class Honours** | Sep 2023 - Sep 2025
University of California, Berkeley, US — MEng Industrial Engineering & Operations Research | Aug 2019 - Dec 2020
Wuhan University, China — BSc Logistics Management | Sep 2015 - Jun 2019